

EBS and EFS

1. Objective

The objective of this practical is to understand and demonstrate AWS storage services:

- Configure and use an Amazon EBS volume.
- Verify EBS data persistence when moving the volume between EC2 instances.
- Study Amazon Elastic File System (EFS).
- Configure an EFS filesystem for shared access between multiple EC2 instances.
- Verify that multiple EC2 instances can access and modify the same EFS data.

Note: The original task mentioned EBS Multi-Attach. For this lab, a normal EBS volume was used to avoid additional cost and to demonstrate EBS persistence through detach and reattach. EFS was used to demonstrate actual simultaneous shared storage.

2. EBS — Elastic Block Store

2.1 Create the EBS Volume

An EBS volume named `additional` was created with:

- Volume type: GP3
- Size: 5 GiB
- Availability Zone: `ap-south-1b`
- Multi-Attach: Disabled

The volume was created in the same Availability Zone as the EC2 instances.

2.2 Attach EBS to the First EC2 Instance

The EBS volume was attached to **servera**.

The device appeared on the instance as:

`/dev/nvme1n1`

The device was checked using:

```
lsblk
```

2.3 Format and Mount the EBS Volume

The volume was formatted with the XFS filesystem:

```
sudo mkfs.xfs /dev/nvme1n1
```

A mount directory was created:

```
sudo mkdir -p /data
```

The volume was mounted:

```
sudo mount /dev/nvme1n1 /data
```

The mount was verified using:

```
df -h /data
```

```
[ec2-user@ip-172-31-4-211 ~]$ df -h /data
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme1n1    5.0G  130M   4.9G   3% /data
[ec2-user@ip-172-31-4-211 ~]$ |
```

2.4 Create Data on EBS

A test file was created:

```
echo "Data created from Instance 1" | sudo tee /data/test.txt
```

The file was verified:

```
cat /data/test.txt
```

```
[ec2-user@ip-172-31-4-211 data]$ ls -lh /data
total 4.0K
-rw-r--r--. 1 root root 29 Sep 17 05:11 test.txt
[ec2-user@ip-172-31-4-211 data]$ cat test.txt
Data created from Instance 1
[ec2-user@ip-172-31-4-211 data]$
```

2.5 Move the EBS Volume to the Second Instance

The EBS volume was:

1. Unmounted from servera.
2. Detached from servera.
3. Attached to serverb.

The volume was then mounted on serverb at /data.

```
[ec2-user@ip-172-31-1-54 ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1      259:0    0   10G  0 disk
├─nvme0n1p1  259:1    0    1M  0 part
├─nvme0n1p2  259:2    0  200M  0 part /boot/efi
└─nvme0n1p3  259:3    0   9.8G  0 part /
nvme1n1      259:4    0    5G  0 disk
[ec2-user@ip-172-31-1-54 ~]$ sudo mkdir /data
[ec2-user@ip-172-31-1-54 ~]$ sudo mount /dev/nvme1n1 /data
[ec2-user@ip-172-31-1-54 ~]$ df -h /data
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme1n1    5.0G  130M  4.9G   3% /data
[ec2-user@ip-172-31-1-54 ~]$ ls -lh /data
total 4.0K
-rw-r--r--. 1 root root 29 Sep 17 05:11 test.txt
```

2.6 Create Additional Data from serverb

A second file was created from serverb:

```
echo "Data created from Instance 2" | sudo tee /data/test2.txt
```

The files were verified:

```
ls -l /data
```

The existing file was checked:

```
cat /data/test.txt
```

The data created from servera was still available.

This confirmed that the data remained available after moving the EBS volume between instances.

```
[ec2-user@ip-172-31-1-54 ~]$ ls -lh /data
total 8.0K
-rw-r--r--. 1 root root 29 Sep 17 05:11 test.txt
-rw-r--r--. 1 root root 29 Sep 17 05:20 test2.txt
[ec2-user@ip-172-31-1-54 ~]$ cat /data/test2.txt
Data created from Instance 2
[ec2-user@ip-172-31-1-54 ~]$ cat /data/test.txt
Data created from Instance 1
[ec2-user@ip-172-31-1-54 ~]$
```

3. Amazon EFS — Elastic File System

3.1 What is EFS?

Amazon Elastic File System (EFS) is a managed, scalable file storage service for AWS.

Unlike a normal EBS volume, EFS is designed to be accessed by multiple EC2 instances at the same time.

EFS uses the **NFS protocol** for mounting the filesystem.

3.2 EBS vs EFS

Feature	EBS	EFS
Storage type	Block storage	File storage
Typical access	Attached to an EC2 instance	Multiple EC2 instances
Availability Zone	AZ-specific	Regional
Protocol	Block device	NFS
Shared access	Not normally shared in this lab	Designed for shared access
Scaling	Volume-based	Automatically scalable
Example use	OS/data disks, databases	Shared web files, application files

4. Create the EFS Filesystem

An EFS filesystem named:

shared-efs

was created in the ap-south-1 Region.

EFS is a **regional filesystem**, so it is not limited to the Availability Zone where the EC2 instance was created.

4.1 EFS Mount Targets

Mount targets were available in:

- ap-south-1a
- ap-south-1b
- ap-south-1c

All mount targets showed **Available**.

The EC2 instances used in the practical were located in:

ap-south-1b

Therefore, they could access EFS through the mount target in that Availability Zone.

5. Configure EFS Security

The EFS security group was configured to allow NFS traffic.

Rule:

Type: NFS

Protocol: TCP

Port: 2049

Source: EC2 security group

This allows the EC2 instances associated with the EC2 security group to communicate with EFS using NFS.

6. Mount EFS on servera

The NFS client package was installed:

```
sudo dnf install -y nfs-utils
```

A mount directory was created:

```
sudo mkdir -p /efs
```

The EFS filesystem was mounted using its DNS name:

```
fs-00c2478acd20e54d4.efs.ap-south-1.amazonaws.com
```

The mount was verified using:

```
df -h /efs
```

The output confirmed that the EFS filesystem was mounted at:

```
/efs
```

```
[ec2-user@ip-172-31-4-211 ~]$ ls -ld /efs
drwxr-xr-x. 2 root root 6 Sep 17 05:45 /efs
[ec2-user@ip-172-31-4-211 ~]$ sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsz=1048576,hard,timeo=600,retrans=2,noresvport fs-00c2478acd20e54d4.efs.ap-south-1.amazonaws.com:/ /efs
[ec2-user@ip-172-31-4-211 ~]$ df -h /efs
Filesystem              Size  Used Avail Use% Mounted on
fs-00c2478acd20e54d4.efs.ap-south-1.amazonaws.com:/ 8.0E  0 8.0E  0% /efs
[ec2-user@ip-172-31-4-211 ~]$ echo "Data created from servera" | sudo tee /efs/servera.txt
Data created from servera
```

7. Create Data from servera

A file was created on the EFS filesystem:

```
echo "Data created from servera" | sudo tee /efs/servera.txt
```

The file was verified:

```
cat /efs/servera.txt
```

8. Mount the Same EFS on serverb

The same EFS filesystem was mounted on **serverb**.

The EFS filesystem was not unmounted from servera.

This is an important difference from the EBS test: both instances were able to access the same EFS filesystem simultaneously.

The mount was verified:

```
df -h /efs
```

9. Verify Shared Data

On serverb, the file created by servera was accessed:

```
cat /efs/servera.txt
```

Output:

Data created from servera

This confirmed that serverb could access data created by servera.

```
[ec2-user@ip-172-31-1-54 ~]$ sudo mkdir -p /efs
[ec2-user@ip-172-31-1-54 ~]$ sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport fs-00c2478acd20e54d4.efs.ap-south-1.amazonaws.com:/ /efs
[ec2-user@ip-172-31-1-54 ~]$ df -h /efs
Filesystem                                Size  Used Avail Use% Mounted on
fs-00c2478acd20e54d4.efs.ap-south-1.amazonaws.com:/ 8.0E  0 8.0E  0% /efs
[ec2-user@ip-172-31-1-54 ~]$ cat /efs/servera.txt
Data created from servera
[ec2-user@ip-172-31-1-54 ~]$ echo "Data created from serverb" | sudo tee /efs/serverb.txt
Data created from serverb
[ec2-user@ip-172-31-1-54 ~]$ cat /efs/serverb.txt
Data created from serverb
[ec2-user@ip-172-31-1-54 ~]$
```

10. Verify Two-Way Access

A file was created from serverb:

```
echo "Data created from serverb" | sudo tee /efs/serverb.txt
```

The file was then accessed from servera:

```
cat /efs/serverb.txt
```

Output:

Data created from serverb

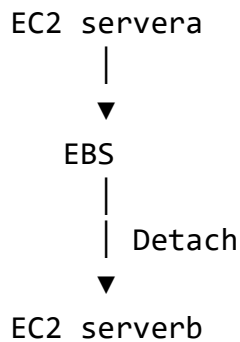
This confirmed that both EC2 instances were accessing the same EFS filesystem.

```
[ec2-user@ip-172-31-1-54 ~]$ sudo mkdir -p /efs
[ec2-user@ip-172-31-1-54 ~]$ sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport fs-00c2478acd20e54d4.efs.ap-south-1.amazonaws.com:/ /efs
[ec2-user@ip-172-31-1-54 ~]$ df -h /efs
Filesystem                                Size  Used Avail Use% Mounted on
fs-00c2478acd20e54d4.efs.ap-south-1.amazonaws.com:/ 8.0E  0 8.0E  0% /efs
[ec2-user@ip-172-31-1-54 ~]$ cat /efs/servera.txt
Data created from servera
[ec2-user@ip-172-31-1-54 ~]$ echo "Data created from serverb" | sudo tee /efs/serverb.txt
Data created from serverb
[ec2-user@ip-172-31-1-54 ~]$ cat /efs/serverb.txt
Data created from serverb
[ec2-user@ip-172-31-1-54 ~]$
```

11. EBS and EFS Practical Comparison

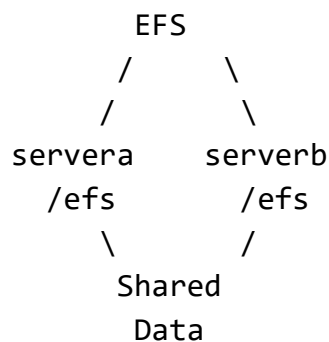
The practical demonstrated two different storage models.

EBS



The volume was moved between instances and its data persisted.

EFS



Both instances accessed the same filesystem at the same time.

Conclusion

This practical provided hands-on experience with AWS block and file storage.

The EBS section demonstrated persistent block storage by creating data on one EC2 instance, detaching the volume, attaching it to another instance, and verifying that the data remained available.

The EFS section demonstrated shared file storage by mounting the same EFS filesystem on two EC2 instances simultaneously. Data created from one instance was

immediately accessible from the other instance, demonstrating the shared nature of EFS.

The practical also covered EFS mount targets, NFS communication, security group configuration, and the difference between Availability Zone-specific EBS storage and regional EFS storage.